OPEN ITEMS CHECKLIST — part 9 (5 November 2026)

Changes since part 8 (30 October 2026). Stages 6j and 6k make the string words optional, stage 6l adds a disassembler, stage 6m adds a decompiler. Source: `forth6809_stage_6m.asm` ; documentation: `ClaudeForth 20261105` ; shadow file: `forth6809 20261105.shd` .

Optional string words (stages 6j, 6k)

- ☑ `STRINGSET` (default 1) assembles the String word set; `STRINGEXT` (default 1) assembles the String extension. Each can be set to 0 with `--define=STRINGSET=0` / `--define=STRINGEXT=0` .
- ☑ `ENVIRONMENT?` answers `STRING` and `STRING-EXT` true only when the matching switch is 1; otherwise it gives the normal false result.
- ☑ `UNITTESTS=1` needs both switches at 1. The source stops with `ERROR "UNITTESTS=1 needs STRINGSET=1 and STRINGEXT=1"` (guard confirmed in lwasm).
- ☑ ANS test file 27 split: **27** (String word set, 50 tests) and **36** (String extension, 14 tests).
- ☑ ROM saved: `STRINGSET=0` 274 bytes, `STRINGEXT=0` 657 bytes, both 931.

Disassembler (stage 6l): `DASM` , `DIS`

- ☑ `DISASM` switch, default 1; default 0 when `UNITTESTS` is nonzero.
- ☑ `DASM ( addr -- addr' )` sends one instruction as text and returns the next address; `DIS ( addr u -- )` lists the instructions that start in addr to addr+u-1, one per line. lwasm-style, hexadecimal only.
- ☑ ANS test file **37_disassembler.tests.fs**: 64 run, 0 failed.
- ☑ Round trip through the assembler: only 4 known `PSHS #$00` differences and 1,083 non-canonical postbytes. Details in shadow `DISASM.1` .

Decompiler (stage 6m): `SEE`

- ☑ `DECOMP` switch, default = `DISASM` . `DECOMP=1` with `DISASM=0` stops with `ERROR "DECOMP=1 needs DISASM=1"` .
- ☑ `SEE ( "name" -- )` rebuilds IF/ELSE/THEN, BEGIN/UNTIL/AGAIN/WHILE/REPEAT, DO/?DO/LOOP/+LOOP, EXIT, the string words and DOES>; CODE words print as `CODE name` with `DASM` lines. `see_check.py` : 19 definitions, 0 differ.
- ☑ Handler labels use the prefix `HND_` (the `H_` prefix is for headers; the first 6m source clashed on `H_DO` , `H_LOOP` , `H_PLUS` , `H_QDO` , `H_EXIT`).
- ☑ The source has a new **SECTION 28** (disassembler and decompiler); the main document lists it as 8.29.

ROM layout (default build, DISASM=1 DECOMP=1)

- ☑ `BASECODE $CEC0` , `BASEDICT $C4F8` , set by conditional EQUs (table in shadow `MEMMAP.20`). 265 headers; `BASEDICT` 2504 bytes; `BASECODE` 12180 bytes polled (ends `$FE54` , 336 spare) and 12499 interrupt-driven (ends `$FF93` , **17 spare**). Unused ROM below `BASEDICT` : 1016 bytes.
- ☑ Cost of the options: `DISASM` +1,789 bytes; `DISASM` + `DECOMP` +2,690 bytes (`DECOMP` alone 901).
- ☑ Globals: 171 bytes used (`GLOBALS_USED = 131+19*DISASM+21*DECOMP`), page zero `$00` - `$AA` .
- ☑ Unit tests: all 19 groups pass with `DISASM=0` (the default when `UNITTESTS` is set). With `DISASM=1 DECOMP=0` groups 0, 2, 5, 13 to 17 fit and pass; with both on only groups 13 and 15 fit (both pass).

Open items

- ☐ **Scripts:** `run all tests.sh` and `build all tests.sh` need the `DISASM` define handled for groups that do not fit (use `DISASM=0`), and 36 and 37 added to the ANS runner `--sections` list.
- ☐ **Interrupt-driven headroom is 17 bytes.** Any further addition needs a lower `BASECODE` / `BASEDICT` .
- ☐ **Not run here:** MAME for the new stages was confirmed by manual testing only; the full `run all tests.sh` in both serial modes with all flag combinations should be repeated.
- ☐ `SEE` limits: `CASE` / `OF` / `ENDOF` are not rebuilt; an `AGAIN` at the end of an `IF` branch is shown as `REPEAT` ; a headerless word shows as `JSR $xxxx` ; output stops after 255 items.

- □ Items still open from part 8 apply (the `HEADER` name-length check; `SEARCH-ORDER` queries in `ENVIRONMENT?` ; README edits).